



Internal Strategic Brief

The Development Intelligence Platform

Behind the meter first. Grid as last resort.

Maximise the closed loop between generation, storage, and consumption

Renewables · BESS · Grid

Data Centers · Port Grids

Simulation · Optimisation · BTM

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01 — Why we need this tool

The project environment has outgrown our current analytical capacity.

The shift in project complexity

- **Projects are no longer single-asset.** We develop stacked systems — renewable generation, BESS, grid interconnection, and co-located consumers — that must be optimised as a system, not as individual components. No spreadsheet does this correctly.
- **Interactions are non-linear.** A change in BESS sizing affects self-sufficiency, which affects grid dependency, which affects connection cost and grid charges, which changes the project economics entirely. These cascades cannot be traced manually.
- **Behind-the-meter logic requires time-series thinking.** Whether the closed loop is serving a data centre or a port, you must model every hour of the year — generation peaks, demand troughs, BESS state of charge — to know how much the grid is actually needed.
- **Grid connection is a cost, not just a revenue route.** Connection costs, use-of-system charges, and reinforcement obligations all increase with connection size. Minimising them is a direct value driver — but only if you can model what size you genuinely need.

The decision-making gap

- **We are making capital allocation decisions on static models** that do not capture the dynamic interaction between generation, storage, and load over time.
- **We have no systematic way to answer the most important development question:** what is the minimum grid connection size that still covers our consumers reliably?
- Without simulation, we default to over-connecting to the grid — which is the most expensive and least competitive outcome.

Root cause

We are designing infrastructure using average assumptions, not time-series simulation. This tool fixes that.

02 — The behind-the-meter principle

This is not a feature. It is the design objective that governs every decision the platform makes.

What 'behind the meter' means in this context

- **The meter is the boundary between NatPower's system and the public grid.** Everything behind it — generation, storage, consumers — belongs to the closed loop. The platform's job is to maximise energy flows within that boundary.
- **The optimisation objective is self-sufficiency, not export revenue.** We want the maximum share of consumer demand covered by own generation and BESS, over every hour of the year. What cannot be covered internally is the residual — and only the residual — that the grid connection needs to handle.
- **The grid connection is sized bottom-up, not top-down.** Traditional developers size grid connections to export generation. We size them to cover what the closed loop genuinely cannot serve — which is typically much smaller, and much cheaper.

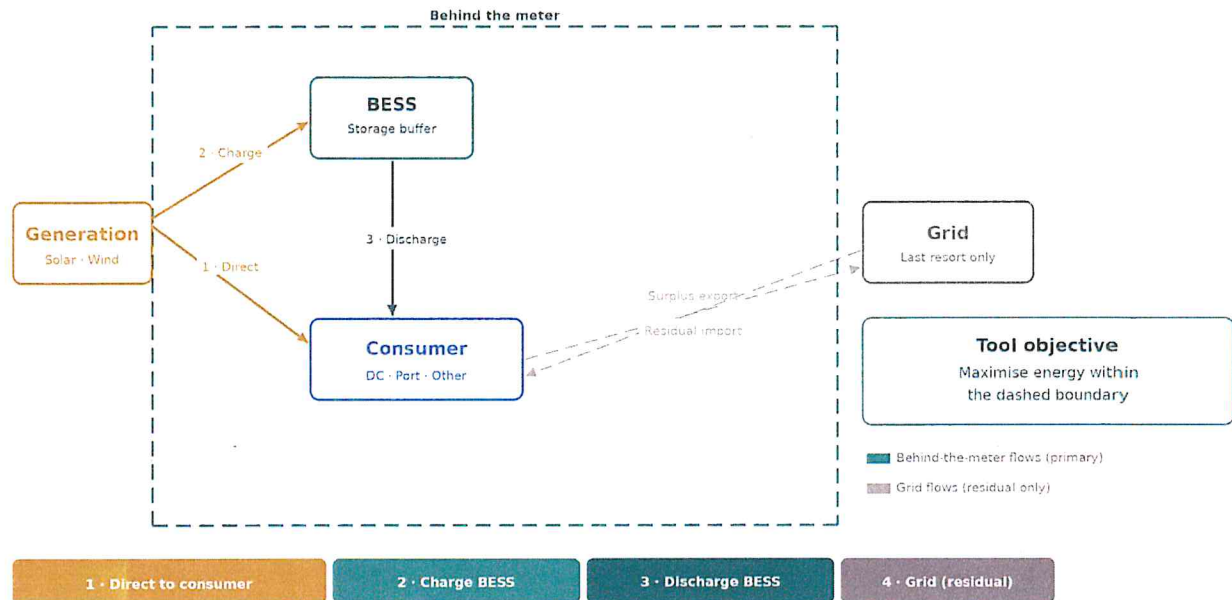
Energy dispatch priority order



Generation flows to the consumer first. Surplus charges the BESS. When generation is insufficient, BESS discharges. The grid only acts when the BESS is depleted and demand still exceeds generation — or when BESS is full and generation surplus has nowhere to go.

What the tool must optimise





Core optimisation problem

Find the combination of generation capacity, BESS size, and grid connection size that achieves a target SSR at minimum total cost. That is the problem. Everything else is inputs and outputs.

03 — What the platform must do

A closed-loop design engine — not a grid-export optimiser.

Core functional requirements

CLOSED-LOOP SIMULATION

- Hourly simulation of generation, BESS, and load over a full year
- BTM dispatch priority logic applied at every time step
- BESS state-of-charge tracked continuously — no static assumptions
- Grid interaction calculated as the residual — not the primary variable

BTM OPTIMISATION ENGINE

- Find the generation + BESS combination that hits a target SSR
- Minimise grid connection size consistent with consumer reliability
- Optimise BESS sizing: more storage delays grid need but costs capex
- Model BESS pre-charge strategies (especially for scheduled loads)

SCENARIO COMPARISON

- Multiple asset configurations run in parallel on the same site
- Compare SCR, SSR, grid dependency, curtailment, and IRR side by side
- Sensitivity sweeps on target SSR, BESS size, generation capacity
- Versioned, shareable scenario results across teams

ECONOMIC OUTPUT

- Capex by asset: generation, BESS, and grid connection
- Opex and avoided grid cost modelling
- Revenue from surplus export and grid services where applicable
- IRR and NPV with SSR and grid size as key sensitivity variables

Design test for every feature

Does this help size the closed loop better, or does it help optimise grid export? If the latter, it belongs in phase 2 — not the MVP.

04 — How the system works

A time-series simulation of the closed loop — with the grid as a boundary condition, not a design variable.

The simulation engine

- **The engine steps through time — hourly — for one or more full years.** At each step it calculates available generation from meteorological data, current consumer demand from load profiles, current BESS state of charge, and applies the BTM dispatch priority to determine energy flows.
- **Grid interaction is computed last, as the residual.** If the closed loop cannot cover demand and the BESS is depleted, the shortfall is the import requirement. If generation exceeds demand and the BESS is full, the surplus is the export. The grid connection only needs to handle the maximum of these residuals.
- **The optimisation layer runs the simulation iteratively** across a range of generation sizes, BESS sizes, and grid connection sizes, evaluating SCR, SSR, cost, and economics at each combination — identifying the Pareto-optimal configurations.

BTM model vs standard energy model

STANDARD ENERGY MODEL

- Grid connection sized to export peak generation
- Revenue = price × exported MWh
- BESS optimised for grid arbitrage
- Consumer load is a cost or a separate model

NATPOWER BTM MODEL

- Grid connection sized to cover residual demand only
- Value = avoided grid cost + consumer off-take revenue
- BESS optimised to maximise SSR and buffer peaks
- Consumer load is the central design constraint

Architectural implication

The consumer load profile is not an input to check against — it is the reference point around which generation and storage are sized. The tool must be designed with the consumer at its centre.

05 — Supported consumer types

Each consumer type has distinct demand characteristics. The BTM logic is shared; the load modelling is specific.

DATA CENTER

- **Demand profile:** continuous, high baseline with variability driven by compute utilisation. Rarely drops to zero. Predictable in shape but not in hourly value.
- **BTM objective:** achieve a target firmness level (e.g. 95% of hours covered by own generation + BESS) — minimising grid dependency while ensuring no compute outage.
- **BESS role:** buffer generation intermittency and smooth demand peaks. Sized to maintain firmness through the longest foreseeable generation gap (e.g. multi-day low-irradiance period).

PORT GRID — COLD IRONING AND VESSEL PROPULSION

- **Demand profile:** intermittent pulses of high magnitude, scheduled in advance. Between vessel arrivals: near-zero port energy demand. During berthing: concentrated MW draw over hours.
- **BTM objective:** pre-charge BESS during low-demand windows — using own generation or off-peak grid — so that vessel arrival demand is covered from storage, avoiding grid peaks entirely.
- **BESS role:** a scheduled energy buffer. Charge before arrival, discharge during cold ironing. The vessel schedule is the dispatch plan — not a reactive signal.
- **Unique optimisation opportunity:** because demand events are known days in advance, the BESS can be pre-charged with precision. This is the highest-value BESS dispatch scenario and is only possible if the platform models the vessel schedule as a first-class input.

FUTURE CONSUMER TYPES

- The platform must be architected to accept new consumer types without rebuilding core logic. Green hydrogen electrolyzers, industrial processes, EV charging hubs, and district heating all follow the same BTM logic with different load profiles.

Multi-consumer requirement

The platform must allow any combination of consumer types on the same site — e.g. a port that also hosts a DC colocation facility. The aggregate load is the sum; the BTM optimisation applies to the combined demand curve.

06 — Strategic value for NatPower

The BTM approach is not just a design philosophy — it is a commercial differentiator.

- **Lower grid connection cost.** A BTM-optimised project may need a fraction of the grid connection a traditional developer would specify. This reduces capex, reduces use-of-system charges, and reduces grid reinforcement risk.
- **Higher revenue per MWh.** Energy sold directly to a co-located consumer commands a premium over grid export. BTM maximises the volume of this direct supply.
- **Reduced grid constraint risk.** In constrained areas, a smaller grid connection is easier to obtain and less exposed to curtailment. BTM design is inherently more resilient to grid capacity limitations.
- **Consumer propositions competitors cannot match.** Offering a DC operator or port authority a genuinely self-sufficient energy system — with defined firmness, known grid backup size, and a fixed off-take tariff — is a fundamentally different commercial proposition from grid-dependent supply.
- **Faster, more defensible investment decisions.** When the BTM simulation shows 85% SSR with a 50 MW grid connection versus 72% SSR with a 100 MW connection, the IC decision is data-driven and auditable — not a judgment call.
- **Port decarbonisation is a growing regulatory opportunity.** EU shore power mandates (AFIR regulation) and port authority procurement are creating structured demand for clean shore power infrastructure. NatPower can participate — but only if we can model and price it correctly.

Competitive position

Most energy developers design generation assets and connect consumers as an afterthought. NatPower designs the consumer relationship first, and sizes the generation and storage system to serve it. This tool is what makes that possible at scale.

07 — Practical use cases

All framed around the BTM optimisation question. These are not hypotheticals — they are decisions we face today.

Ref	Title	Simulation question answered
BTM-01	What grid connection do I actually need?	Given a consumer load profile, site generation potential, and a target SSR (e.g. 85%), what is the minimum grid connection size that ensures consumer reliability?
BTM-02	BESS sizing for SSR improvement	How many MWh of BESS turns a 70% SSR into a 90% SSR? At what point does adding storage deliver diminishing returns versus enlarging the grid connection?
DC-01	Self-sufficient supply for a Data Center	Design a generation + BESS system for a 100 MW DC. Determine optimal solar/wind mix, BESS size, and minimum grid connection to achieve 95% firmness.
PORT-01	Pre-charge BESS for vessel schedule	Import vessel schedule. Identify cold ironing demand events. Size BESS to cover peak berthing day from storage alone. Pre-charge windows filled by site generation.
PORT-02	Avoid grid reinforcement at congested port	Port grid is constrained at 30 MW. Vessel demand peaks at 80 MW. Model BESS as a grid buffer — prove reinforcement is unnecessary with the right BESS size.
BTM-03	Multi-consumer site: DC + Port colocation	One site serves both a DC (continuous baseline) and a port (scheduled peaks). Find the BESS sizing that smooths both profiles and minimises the combined grid connection.
BTM-04	Grid-constrained site — BTM unlocks the project	400 MW generation potential, 50 MW grid capacity. Standard model: unviable. BTM model: co-locate a 200 MW DC, size BESS to absorb generation peaks. Project viable.
FIN-01	SSR sensitivity for financing	Lender requires 90% SSR covenant. Define the BESS and generation combination that meets the covenant with minimum capex. Present as the bankable configuration.

08 — Product delivery priorities

MVP is the BTM simulation engine. Everything else follows once that core works.

MVP — PHASE 1

- **BTM dispatch logic engine (priority order hardcoded)**
- **SCR and SSR calculation at every time step**
- **Residual grid import/export as derived output**
- Hourly simulation, full year, from meteorological data
- Solar and wind generation models
- BESS model with state-of-charge tracking
- Grid connection as boundary condition
- DC load profile input (continuous baseline)
- Port load module: vessel schedule → hourly demand curve
- Scenario comparison: SCR, SSR, grid size, IRR
- Web UI usable by development managers, not just engineers

PHASE 2 / PHASE 3

- Automated BTM optimisation (parametric sweep to hit target SSR)
- Schedule-aware BESS pre-charge for port loads
- Multi-consumer aggregate demand modelling
- BESS degradation over project life
- Live grid constraint data integration
- Dynamic price inputs and arbitrage logic
- Extensible consumer type framework (hydrogen, EV, industrial)
- Multi-site portfolio view
- Lender-grade SSR covenant reporting
- API connectivity to port management / DC CIM systems

MVP definition test

A development manager must be able to answer 'what grid connection do I need for this site and this consumer?' using only Phase 1 features. If that question cannot be answered from Phase 1, the MVP is incomplete.

Open items requiring definition before build

- **Target SSR levels by project type.** What does 'reliable' mean for DC vs port vs grid backup? This number drives BESS sizing fundamentally and needs an owner.
- **Vessel class library.** Standard cold ironing MW draw by vessel type: container, cruise, bulk, RoRo, tanker.
- **BESS round-trip efficiency and degradation assumptions** for lifecycle modelling.
- **Grid import tariff structure.** Needed to value avoided grid cost correctly in the economic model.

09 — CEO expectations from the Digital team

This is a strategic build — not a technology experiment.

The standard I am setting

This platform will be used to make investment decisions worth hundreds of millions of euros and to structure commercial agreements with data centre operators and port authorities. Accuracy on SCR and SSR calculations is non-negotiable — these are the numbers that go into contracts and investment committee packs.

Specific demands

- **The BTM dispatch logic must be the engine's foundation — not a feature added later.** Every simulation must apply the priority order: direct consumption → BESS charge → BESS discharge → grid. This logic must be transparent, auditable, and correct at every time step.
- **SCR and SSR must be first-class outputs.** Not buried in a table. Not derived manually. These are the two primary metrics by which every project configuration is evaluated — prominent, immediate, and unambiguous in the interface.
- **The grid connection size must be an output, not an input.** At minimum, the platform must answer: 'given this consumer profile, this generation, and this BESS — here is the minimum grid connection you need.' That is the core question the tool exists to answer.
- **Do not build a visualisation tool.** Charts are outputs. Calculation chains are the product. Every number must be traceable to its model assumption.
- **Design for the development manager, not the engineer.** If it takes more than 30 minutes to configure and run a scenario, rebuild the interface.
- **Propose technology architecture within 30 days.** Build vs adapt a commercial platform vs hybrid. Present trade-offs with honest long-term maintenance cost estimates included.
- **Deliver a working MVP in six months.** A real scenario, on a real site, producing SCR/SSR/grid size outputs that go into a real investment committee pack.

Final word

NatPower's differentiation is the ability to offer consumers — data centres, ports, and others — a genuinely self-sufficient energy system rather than grid-dependent supply. This tool is the foundation of that proposition. Build it like it matters — because it does.